

Numerical Examples – Unit 1

Example 1

The pdf of a random variable X is given by

$$f_X(x) = \begin{cases} k, & a \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

where k is a constant.

(a) Determine the value of k .

(b) Let $a = -1$ and $b = 2$. Calculate $P(|X| \leq c)$ for $c = \frac{1}{2}$.

Solution

(a) From property 1 of $f_X(x)$, k must be a positive constant. From property 2 of a pdf,

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_a^b k dx = k(b - a) = 1.$$

Hence,

$$k = \frac{1}{b - a}.$$

Thus,

$$f_X(x) = \begin{cases} \frac{1}{b - a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

A random variable X having the above pdf is called a *uniform random variable*.

(b) With $a = -1$ and $b = 2$,

$$f_X(x) = \begin{cases} \frac{1}{3}, & -1 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\begin{aligned} P\left(|X| \leq \frac{1}{2}\right) &= P\left(-\frac{1}{2} \leq X \leq \frac{1}{2}\right) \\ &= \int_{-1/2}^{1/2} f_X(x) dx \\ &= \int_{-1/2}^{1/2} \frac{1}{3} dx \\ &= \frac{1}{3}. \end{aligned}$$

Example 2

The joint pdf of X and Y is given by

$$f_{XY}(x, y) = xy e^{-(x^2+y^2)/2} u(x)u(y).$$

- (a) Find the marginal pdfs $f_X(x)$ and $f_Y(y)$.
- (b) Are X and Y independent?

Solution

(a) The marginal pdf of X is

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dy \\ &= \int_0^{\infty} xy e^{-(x^2+y^2)/2} u(x) dy \\ &= x e^{-x^2/2} u(x) \int_0^{\infty} y e^{-y^2/2} dy \\ &= x e^{-x^2/2} u(x). \end{aligned}$$

Since $f_{XY}(x, y)$ is symmetric in x and y ,

$$f_Y(y) = y e^{-y^2/2} u(y).$$

(b) Since

$$f_{XY}(x, y) = f_X(x) f_Y(y),$$

we conclude that X and Y are independent.

Example 3

The input to a noisy communication channel is a binary random variable X with

$$P(X = 0) = P(X = 1) = \frac{1}{2}.$$

The output of the channel Z is given by

$$Z = X + Y,$$

where Y is the additive noise introduced by the channel. Assuming that X and Y are independent and $Y = N(0, 1)$, find the density function of Z .

Solution

Using the law of total probability,

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(Z \leq z \mid X = 0)P(X = 0) + P(Z \leq z \mid X = 1)P(X = 1). \end{aligned}$$

Since $Z = X + Y$,

$$\begin{aligned} P(Z \leq z \mid X = 0) &= P(X + Y \leq z \mid X = 0) \\ &= P(Y \leq z) \\ &= F_Y(z). \end{aligned}$$

Similarly,

$$\begin{aligned} P(Z \leq z \mid X = 1) &= P(X + Y \leq z \mid X = 1) \\ &= P(Y \leq z - 1) \\ &= F_Y(z - 1). \end{aligned}$$

Hence,

$$F_Z(z) = \frac{1}{2}F_Y(z) + \frac{1}{2}F_Y(z - 1).$$

Since $Y \sim N(0, 1)$,

$$f_Y(y) = \frac{1}{\sqrt{2\pi}}e^{-y^2/2}.$$

Therefore,

$$\begin{aligned} f_Z(z) &= \frac{dF_Z(z)}{dz} \\ &= \frac{1}{2}f_Y(z) + \frac{1}{2}f_Y(z - 1) \\ &= \frac{1}{2} \left[\frac{1}{\sqrt{2\pi}}e^{-z^2/2} + \frac{1}{\sqrt{2\pi}}e^{-(z-1)^2/2} \right]. \end{aligned}$$

Example 4

Find the covariance of X and Y if

- (a) they are independent, and
- (b) Y is related to X by $Y = aX + b$.

Solution

(a) If X and Y are independent, then

$$\begin{aligned}\sigma_{XY} &= E[XY] - E[X]E[Y] \\ &= E[X]E[Y] - E[X]E[Y] \\ &= 0.\end{aligned}$$

(b) Let $Y = aX + b$.

$$\begin{aligned}E[XY] &= E[X(aX + b)] \\ &= aE[X^2] + bE[X] \\ &= aE[X^2] + b\mu_X,\end{aligned}$$

and

$$\begin{aligned}\mu_Y &= E[Y] \\ &= E[aX + b] \\ &= aE[X] + b \\ &= a\mu_X + b.\end{aligned}$$

Thus,

$$\begin{aligned}\sigma_{XY} &= E[XY] - E[X]E[Y] \\ &= aE[X^2] + b\mu_X - \mu_X(a\mu_X + b) \\ &= a(E[X^2] - \mu_X^2) \\ &= a\sigma_X^2.\end{aligned}$$

Note: The result of part (a) states that if X and Y are independent, then they are uncorrelated. The converse is not necessarily true.

Example 5

Let X and Y be defined by

$$X = \cos \theta, \quad Y = \sin \theta$$

where θ is a random variable uniformly distributed over $[0, 2\pi]$.

- (a) Show that X and Y are uncorrelated.
- (b) Show that X and Y are not independent.

Solution

Since θ is uniformly distributed over $[0, 2\pi]$,

$$f_{\theta}(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 < \theta < 2\pi, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Showing that X and Y are uncorrelated

Using expectations,

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^{2\pi} \cos \theta f_{\theta}(\theta) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \cos \theta d\theta \\ &= 0. \end{aligned}$$

Similarly,

$$E[Y] = \frac{1}{2\pi} \int_0^{2\pi} \sin \theta d\theta = 0.$$

Also,

$$\begin{aligned} E[XY] &= \frac{1}{2\pi} \int_0^{2\pi} \cos \theta \sin \theta d\theta \\ &= \frac{1}{4\pi} \int_0^{2\pi} \sin 2\theta d\theta \\ &= 0. \end{aligned}$$

Hence,

$$E[XY] = E[X]E[Y].$$

Therefore, X and Y are **uncorrelated**.

(b) Showing that X and Y are not independent

$$\begin{aligned} E[X^2] &= \frac{1}{2\pi} \int_0^{2\pi} \cos^2 \theta d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2}(1 + \cos 2\theta) d\theta \\ &= \frac{1}{2}, \end{aligned}$$

and similarly,

$$E[Y^2] = \frac{1}{2}.$$

Now,

$$\begin{aligned} E[X^2 Y^2] &= \frac{1}{2\pi} \int_0^{2\pi} \cos^2 \theta \sin^2 \theta d\theta \\ &= \frac{1}{8\pi} \int_0^{2\pi} (1 - \cos 4\theta) d\theta \\ &= \frac{1}{8}. \end{aligned}$$

But,

$$E[X^2]E[Y^2] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Since

$$E[X^2 Y^2] \neq E[X^2]E[Y^2],$$

X and Y are **not independent**.

Hence, X and Y are uncorrelated but not independent.

Example 6

Consider a random process $X(t)$ given by

$$X(t) = A \cos(\omega t + \Theta),$$

where A and ω are constants and Θ is a uniform random variable over $[-\pi, \pi]$. Show that $X(t)$ is wide-sense stationary (WSS).

Solution

Since Θ is uniformly distributed over $[-\pi, \pi]$,

$$f_{\Theta}(\theta) = \begin{cases} \frac{1}{2\pi}, & -\pi \leq \theta \leq \pi, \\ 0, & \text{otherwise.} \end{cases}$$

Mean of the process

$$\begin{aligned} \mu_X(t) &= E[X(t)] \\ &= \int_{-\infty}^{\infty} A \cos(\omega t + \theta) f_{\Theta}(\theta) d\theta \\ &= \frac{A}{2\pi} \int_{-\pi}^{\pi} \cos(\omega t + \theta) d\theta \\ &= 0. \end{aligned}$$

Hence, the mean is a constant.

Autocorrelation function

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \cos(\omega t + \theta) \cos(\omega(t + \tau) + \theta) d\theta. \end{aligned}$$

Using

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)],$$

we obtain

$$\begin{aligned} R_{XX}(t, t + \tau) &= \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \frac{1}{2} [\cos(\omega\tau) + \cos(2\omega t + 2\theta + \omega\tau)] d\theta \\ &= \frac{A^2}{2} \cos(\omega\tau). \end{aligned}$$

Since the mean of $X(t)$ is constant and the autocorrelation depends only on the time difference τ ,

$$R_{XX}(t, t + \tau) = R_{XX}(\tau),$$

we conclude that $X(t)$ is a **wide-sense stationary (WSS)** process.

Note: $R_{XX}(\tau)$ is periodic with period

$$T_0 = \frac{2\pi}{\omega}.$$

A WSS random process is called periodic if its autocorrelation function is periodic.

Example 7

Consider a random process $X(t)$ given by

$$X(t) = A \cos(\omega t + \theta),$$

where ω and θ are constants and A is a random variable. Determine whether $X(t)$ is WSS.

Solution

The mean value is

$$\begin{aligned} \mu_X(t) &= E[X(t)] \\ &= E[A \cos(\omega t + \theta)] \\ &= \cos(\omega t + \theta) E[A]. \end{aligned}$$

Hence, the mean of $X(t)$ is not constant unless

$$E[A] = 0.$$

Next, the autocorrelation function is

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[A^2 \cos(\omega t + \theta) \cos(\omega(t + \tau) + \theta)] \\ &= \frac{1}{2} \left[\cos(\omega\tau) + \cos(2\omega t + 2\theta + \omega\tau) \right] E[A^2]. \end{aligned}$$

Since the autocorrelation depends on both t and τ , it is not a function of the time difference alone.

Conclusion: The process $X(t)$ is **not WSS**.

Example 8

Consider a random process $X(t)$ given by

$$X(t) = A \cos \omega t + B \sin \omega t,$$

where ω is constant and A and B are random variables.

(a) Show that the condition

$$E[A] = E[B] = 0$$

is necessary for $X(t)$ to be stationary.

(b) Show that $X(t)$ is WSS if and only if the random variables A and B are uncorrelated with equal variance, i.e.,

$$E[AB] = 0, \quad E[A^2] = E[B^2] = \sigma^2.$$

Solution

(a) The mean of the process is

$$\begin{aligned} \mu_X(t) &= E[X(t)] \\ &= E[A] \cos \omega t + E[B] \sin \omega t. \end{aligned}$$

For stationarity, $\mu_X(t)$ must be independent of t , which is possible only if

$$E[A] = E[B] = 0.$$

(b) If $X(t)$ is WSS, then

$$E[X^2(0)] = E\left[X^2\left(\frac{\pi}{2\omega}\right)\right] = R_{XX}(0) = \sigma_X^2.$$

But

$$X(0) = A, \quad X\left(\frac{\pi}{2\omega}\right) = B.$$

Hence,

$$E[A^2] = E[B^2] = \sigma_X^2 = \sigma^2.$$

Now,

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[(A \cos \omega t + B \sin \omega t)(A \cos \omega(t + \tau) + B \sin \omega(t + \tau))] \\ &= \sigma^2 \cos \omega \tau + E[AB] \sin(2\omega t + \omega \tau). \end{aligned}$$

This will be a function of τ only if

$$E[AB] = 0.$$

Conversely, if

$$E[AB] = 0, \quad E[A^2] = E[B^2] = \sigma^2,$$

then

$$\mu_X(t) = 0, \quad R_{XX}(t, t + \tau) = \sigma^2 \cos \omega \tau = R_{XX}(\tau).$$

Hence, $X(t)$ is **WSS**.

Example 9

A random process $X(t)$ is said to be *covariance-stationary* if the covariance of $X(t)$ depends only on the time difference

$$\tau = t_2 - t_1,$$

that is,

$$C_{XX}(t, t + \tau) = C_{XX}(\tau).$$

Let

$$X(t) = (A + 1) \cos t + B \sin t,$$

where A and B are independent random variables for which

$$E[A] = E[B] = 0, \quad E[A^2] = E[B^2] = 1.$$

Show that $X(t)$ is not WSS, but it is covariance-stationary.

Solution

Mean of the process

$$\begin{aligned} \mu_X(t) &= E[X(t)] \\ &= E[(A + 1) \cos t + B \sin t] \\ &= E[A + 1] \cos t + E[B] \sin t \\ &= \cos t. \end{aligned}$$

Since the mean depends on t , the process $X(t)$ is **not WSS**.

Autocorrelation

$$\begin{aligned}
 R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] \\
 &= E[((A+1)\cos t_1 + B\sin t_1)((A+1)\cos t_2 + B\sin t_2)] \\
 &= E[(A+1)^2]\cos t_1\cos t_2 + E[B^2]\sin t_1\sin t_2 \\
 &\quad + E[(A+1)B](\cos t_1\sin t_2 + \sin t_1\cos t_2).
 \end{aligned}$$

Now,

$$\begin{aligned}
 E[(A+1)^2] &= E[A^2 + 2A + 1] = E[A^2] + 2E[A] + 1 = 2, \\
 E[(A+1)B] &= E[AB] + E[B] = E[A]E[B] + E[B] = 0, \\
 E[B^2] &= 1.
 \end{aligned}$$

Substituting,

$$\begin{aligned}
 R_{XX}(t_1, t_2) &= 2\cos t_1\cos t_2 + \sin t_1\sin t_2 \\
 &= \cos(t_2 - t_1) + \cos t_1\cos t_2.
 \end{aligned}$$

Covariance function

$$\begin{aligned}
 C_{XX}(t_1, t_2) &= R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) \\
 &= \cos(t_2 - t_1) + \cos t_1\cos t_2 - \cos t_1\cos t_2 \\
 &= \cos(t_2 - t_1).
 \end{aligned}$$

Hence,

$$C_{XX}(t_1, t_2) = C_{XX}(t_2 - t_1),$$

which depends only on the time difference.

Conclusion:

- $X(t)$ is **not WSS** (mean depends on t),
- but $X(t)$ is **covariance-stationary**.

Example 10

Show that the process $X(t)$ defined by

$$X(t) = A\cos(\omega t + \Theta)$$

is ergodic in both the mean and the autocorrelation.

Solution

Time-average mean

From the definition of time average,

$$\begin{aligned}\bar{x} &= \langle x(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} A \cos(\omega t + \theta) dt \\ &= \frac{A}{T_0} \int_{-T_0/2}^{T_0/2} \cos(\omega t + \theta) dt \\ &= 0,\end{aligned}$$

where

$$T_0 = \frac{2\pi}{\omega}.$$

Time-average autocorrelation

$$\begin{aligned}\bar{R}_{XX}(\tau) &= \langle x(t)x(t+\tau) \rangle \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} A^2 \cos(\omega t + \theta) \cos[\omega(t+\tau) + \theta] dt \\ &= \frac{A^2}{T_0} \int_{-T_0/2}^{T_0/2} \frac{1}{2} [\cos \omega \tau + \cos(2\omega t + 2\theta + \omega \tau)] dt \\ &= \frac{A^2}{2} \cos \omega \tau.\end{aligned}$$

Thus,

$$\mu_X(t) = E[X(t)] = \bar{x},$$

and

$$R_{XX}(\tau) = E[X(t)X(t+\tau)] = \bar{R}_{XX}(\tau).$$

Hence, $X(t)$ is **ergodic in both the mean and the autocorrelation**.

Example 11

A class of modulated random signal $Y(t)$ is defined by

$$Y(t) = AX(t) \cos(\omega_c t + \Theta),$$

where $X(t)$ is the random message signal and $A \cos(\omega_c t + \Theta)$ is the carrier.

The random message signal $X(t)$ is a zero-mean stationary random process with autocorrelation $R_{XX}(\tau)$ and power spectrum $S_{XX}(\omega)$. The carrier amplitude A and the frequency ω_c are constants, and phase Θ is uniformly distributed over $[0, 2\pi]$. Assuming $X(t)$ and Θ are independent, find the mean, autocorrelation, and power spectrum of $Y(t)$.

Solution

Mean

$$\begin{aligned}\mu_Y(t) &= E[Y(t)] \\ &= E[AX(t) \cos(\omega_c t + \Theta)] \\ &= AE[X(t)]E[\cos(\omega_c t + \Theta)] \\ &= 0,\end{aligned}$$

since $E[X(t)] = 0$ and $X(t)$ and Θ are independent.

Autocorrelation

$$\begin{aligned}R_{YY}(t, t + \tau) &= E[Y(t)Y(t + \tau)] \\ &= E[A^2 X(t)X(t + \tau) \cos(\omega_c t + \Theta) \cos(\omega_c(t + \tau) + \Theta)] \\ &= \frac{A^2}{2} E[X(t)X(t + \tau)] E[\cos \omega_c \tau + \cos(2\omega_c t + \omega_c \tau + 2\Theta)] \\ &= \frac{A^2}{2} R_{XX}(\tau) \cos \omega_c \tau.\end{aligned}$$

Hence,

$$R_{YY}(t, t + \tau) = R_{YY}(\tau).$$

Since the mean is constant and autocorrelation depends only on τ , $Y(t)$ is WSS.

Power spectral density

$$S_{YY}(\omega) = \mathcal{F}\{R_{YY}(\tau)\} = \frac{A^2}{2} \mathcal{F}\{R_{XX}(\tau) \cos \omega_c \tau\}.$$

Using

$$\mathcal{F}\{R_{XX}(\tau)\} = S_{XX}(\omega),$$

and

$$\mathcal{F}\{\cos \omega_c \tau\} = \pi\delta(\omega - \omega_c) + \pi\delta(\omega + \omega_c),$$

we obtain

$$\begin{aligned}S_{YY}(\omega) &= \frac{A^2}{4} S_{XX}(\omega) * [\pi\delta(\omega - \omega_c) + \pi\delta(\omega + \omega_c)] \\ &= \frac{A^2}{4} [S_{XX}(\omega - \omega_c) + S_{XX}(\omega + \omega_c)].\end{aligned}$$

Example 12

Let $X(t)$ and $Y(t)$ be both zero-mean WSS random processes. Define

$$Z(t) = X(t) + Y(t).$$

- (a) Determine the autocorrelation and power spectrum of $Z(t)$ if $X(t)$ and $Y(t)$ are jointly WSS.
- (b) Repeat part (a) if $X(t)$ and $Y(t)$ are orthogonal.
- (c) Show that if $X(t)$ and $Y(t)$ are orthogonal, then the mean square of $Z(t)$ equals the sum of the mean squares of $X(t)$ and $Y(t)$.

Solution

(a) Autocorrelation

$$\begin{aligned} R_{ZZ}(t_1, t_2) &= E[Z(t_1)Z(t_2)] \\ &= R_{XX}(t_1, t_2) + R_{XY}(t_1, t_2) + R_{YX}(t_1, t_2) + R_{YY}(t_1, t_2). \end{aligned}$$

If jointly WSS,

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{XY}(\tau) + R_{YX}(\tau) + R_{YY}(\tau).$$

Taking Fourier transforms,

$$S_{ZZ}(\omega) = S_{XX}(\omega) + S_{XY}(\omega) + S_{YX}(\omega) + S_{YY}(\omega).$$

(b) Orthogonal case

If $X(t)$ and $Y(t)$ are orthogonal,

$$R_{XY}(\tau) = R_{YX}(\tau) = 0.$$

Hence,

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau),$$

and

$$S_{ZZ}(\omega) = S_{XX}(\omega) + S_{YY}(\omega).$$

(c) Mean square

$$R_{ZZ}(0) = R_{XX}(0) + R_{YY}(0),$$

or

$$E[Z^2(t)] = E[X^2(t)] + E[Y^2(t)].$$

Example 13

Two random processes are

$$X(t) = A \cos(\omega t + \Theta), \quad Y(t) = A \sin(\omega t + \Theta),$$

where A and ω are constants and Θ is uniformly distributed over $[0, 2\pi]$.

Find the cross-correlation of $X(t)$ and $Y(t)$.

Solution

$$\begin{aligned} R_{XY}(t, t + \tau) &= E[X(t)Y(t + \tau)] \\ &= E[A^2 \cos(\omega t + \Theta) \sin(\omega(t + \tau) + \Theta)] \\ &= \frac{A^2}{2} \sin \omega \tau. \end{aligned}$$

Similarly,

$$R_{YX}(t, t + \tau) = -\frac{A^2}{2} \sin \omega \tau.$$

Hence,

$$R_{XY}(-\tau) = R_{YX}(\tau).$$

Example 14

Let

$$X(t) = A \cos \omega t + B \sin \omega t,$$

$$Y(t) = B \cos \omega t - A \sin \omega t,$$

where ω is constant and A and B are independent random variables with zero mean and variance σ^2 .

Find the cross-correlation of $X(t)$ and $Y(t)$.

Solution

$$\begin{aligned} R_{XY}(t_1, t_2) &= E[X(t_1)Y(t_2)] \\ &= E[(A \cos \omega t_1 + B \sin \omega t_1)(B \cos \omega t_2 - A \sin \omega t_2)]. \end{aligned}$$

Since

$$E[AB] = 0, \quad E[A^2] = E[B^2] = \sigma^2,$$

we obtain

$$\begin{aligned} R_{XY}(t_1, t_2) &= \sigma^2(\sin \omega t_1 \cos \omega t_2 - \cos \omega t_1 \sin \omega t_2) \\ &= \sigma^2 \sin[\omega(t_1 - t_2)]. \end{aligned}$$

Letting $\tau = t_2 - t_1$,

$$R_{XY}(\tau) = -\sigma^2 \sin \omega \tau.$$

Example 15

A WSS random process $X(t)$ is applied to the input of an LTI system with impulse response

$$h(t) = 3e^{-2t}u(t).$$

Find the mean value of the output $Y(t)$ if

$$E[X(t)] = 2.$$

Solution

The frequency response of the system is

$$H(\omega) = \mathcal{F}\{h(t)\} = \frac{3}{j\omega + 2}.$$

For a WSS input to an LTI system,

$$\mu_Y = \mu_X H(0).$$

Hence,

$$\mu_Y = 2 \left(\frac{3}{2} \right) = 3.$$

Example 16

The input $X(t)$ to the RC filter shown in Figure. 1 is a white noise process.

- (a) Determine the power spectrum of the output process $Y(t)$.
- (b) Determine the autocorrelation and mean-square value of $Y(t)$.

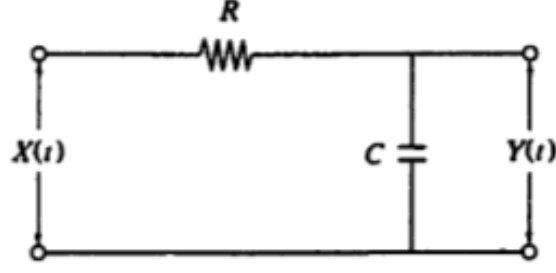


Figure 1: RC Filter

Solution

The frequency response of the RC filter is

$$H(\omega) = \frac{1}{1 + j\omega RC}.$$

Since the input is white noise,

$$S_{XX}(\omega) = \frac{\eta}{2}.$$

(a) Output power spectrum

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= \frac{1}{1 + (\omega RC)^2} \frac{\eta}{2}. \end{aligned}$$

(b) Autocorrelation

Rewriting,

$$S_{YY}(\omega) = \frac{\eta}{2} \frac{1}{2RC} \frac{2(1/RC)}{\omega^2 + (1/RC)^2}.$$

Using the Fourier transform pair,

$$\mathcal{F}^{-1} \left\{ \frac{2a}{\omega^2 + a^2} \right\} = e^{-a|\tau|},$$

we obtain

$$R_{YY}(\tau) = \frac{\eta}{4RC} e^{-|\tau|/(RC)}.$$

The mean-square value is

$$E[Y^2(t)] = R_{YY}(0) = \frac{\eta}{4RC}.$$

Example 17

The input $X(t)$ to an ideal bandpass filter with frequency response shown in Figure. 2 is a white noise process. Determine the total noise power at the output of the filter.

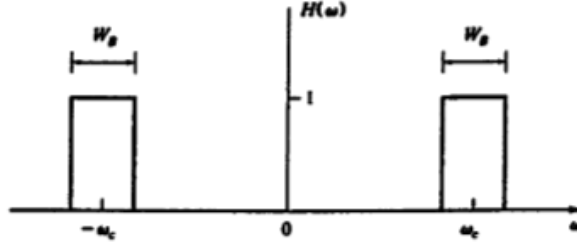


Figure 2: Bandpass filter response

Solution

For white noise,

$$S_{XX}(\omega) = \frac{\eta}{2}.$$

The output power spectrum is

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega) = \frac{\eta}{2} |H(\omega)|^2.$$

The total noise power at the output is

$$\begin{aligned} E[Y^2(t)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\omega) d\omega \\ &= \frac{\eta}{2} \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(\omega)|^2 d\omega. \end{aligned}$$

For an ideal bandpass filter of bandwidth W_B ,

$$E[Y^2(t)] = \frac{\eta}{2} \frac{1}{2\pi} (2W_B) = \eta B,$$

where

$$B = \frac{W_B}{2\pi} \quad (\text{Hz}).$$